

all the instances that you want to launch, which is both time taking and error-prone.

**Immutable VM:** In this approach, you build your own custom AMI with all the special needs and then launch the instance. Once the custom AMI is ready, you can launch any number of instances quickly without any intervention. If you need a VM with another set of requirements, you repeat the process of building another custom AMI and launch the instances. AWS offers special tooling such as AWS CLI for building custom VMs.

So far so good, as long as you want all your computing infrastructure only on AWS. However, AWS is not the only player in this market. Other cloud providers like Google Cloud, Azure Cloud, etc, also offer images to launch VMs on their cloud. Your organisation may want to deploy different apps on different clouds.

In such a case, your DevOps team needs to use different tools to build machine images for different clouds. Though it is not impossible to live in such an environment, it definitely calls for a uniform way and toolset to build identical images for different cloud platforms. That's where the Packer tool from HashiCorp comes into the picture.

According to their website, Packer is an open source tool that lets you create identical machine images for multiple platforms from a single source template. All you need is to develop a Packer template using their HashiCorp Configuration Language (HCL) and ask the Packer tool to build an AMI based on the template. In other words, Packer helps you in practising IaC or Infrastructure as a Code.

This article explains the way to build an image using the Packer tool.

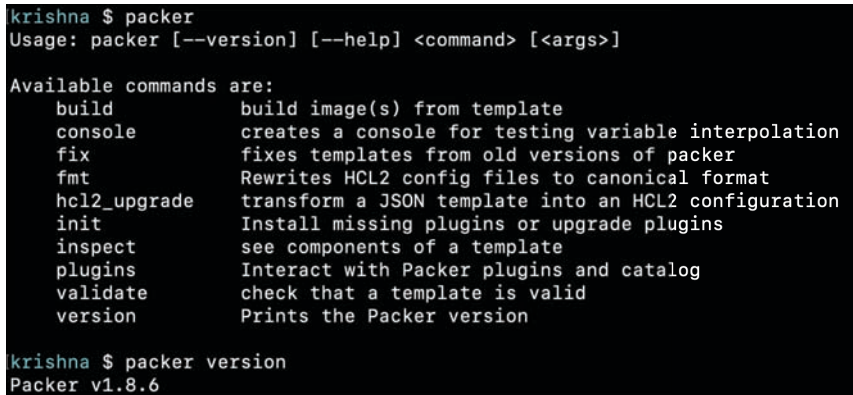
## Installing Packer

The Packer tool is available on most of the popular platforms, and installing it is easy and simple.

The following commands install Packer on any Linux platform:

```
curl -fsSL https://apt.releases.hashicorp.com/gpg | sudo apt-key add -
sudo apt-add-repository "deb [arch=amd64] https://apt.releases.hashicorp.com $(lsb_release -cs) main"
sudo apt-get update
sudo apt-get install packer
```

In case your DevOps team is using Mac, then the



```
krishna $ packer
Usage: packer [--version] [--help] <command> [<args>]

Available commands are:
  build      build image(s) from template
  console    creates a console for testing variable interpolation
  fix        fixes templates from old versions of packer
  fmt        Rewrites HCL2 config files to canonical format
  hcl2_upgrade transform a JSON template into an HCL2 configuration
  init       Install missing plugins or upgrade plugins
  inspect    see components of a template
  plugins    Interact with Packer plugins and catalog
  validate   check that a template is valid
  version    Prints the Packer version

krishna $ packer version
Packer v1.8.6
```

Figure 1: Installing Packer

Homebrew can be used to install Packer. All you need is to run the following commands after installing the latest version of Homebrew.

```
brew tap hashicorp/tap
brew install hashicorp/tap/packer
```

Once Packer is installed, you can verify if the installation is proper by running the following command on the command prompt:

```
packer
```

The result of running the above commands appears in Figure 1. It suggests the correct syntax and lists the available commands.

You may want to try the easiest command:

```
packer version
```

In my case, I found that Packer version 1.8.6 is running on my Mac.

## Building an AMI using Packer

To demonstrate the Packer tool, let us build an AWS machine image or AMI. AWS offers a free-tier account, using which you can build AMIs and launch small VMs.

As the first step, visit <https://console.aws.amazon.com/> to create an AWS account and make a note of the security credentials. Packer communicates with AWS using these credentials over SSH. You may either specify the credentials in the Packer template or store them on the machine, which Packer can locate whenever it needs.

Obviously, the latter approach is much more convenient and secure. To this end, use the AWS CLI to store the AWS security credentials securely. For example, use the following command on a Mac to install AWS CLI:

```
brew install awscli
```